5 **(a)**

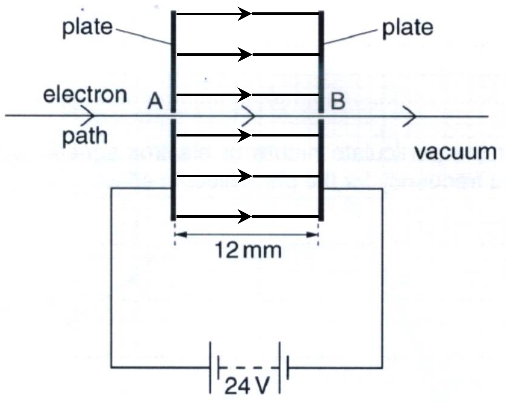

Fig. 5.1

(b) **(i)** $F = eE = e\dfrac{V}{d} = 1.60\times10^{-19}\times\dfrac{24}{12\times10^{-3}} = 3.2\times10^{-16}$ N

(ii) Work done by the force $F = Fd = \left(-3.2\times10^{-16}\right)\left(12\times10^{-3}\right) = -3.84\times10^{-18}$ J

(because F points towards plate A, in the opposite direction to the displacement from A to B)

Magnitude of work done = 3.8×10^{-18} J

(iii) By conservation of energy,

$KE_{\text{A}} - W.D.$ by $F = KE_{\text{B}}$

$$\frac{1}{2}m_{\text{e}}u^2 - W.D. = \frac{1}{2}m_{\text{e}}v^2$$

$$\frac{1}{2}\left(9.11\times10^{-31}\right)\left(4.5\times10^{6}\right)^2 - 3.84\times10^{-18} = \frac{1}{2}\left(9.11\times10^{-31}\right)v^2$$

$$\therefore v = 3.45\times10^{6}\ \text{m s}^{-1}$$

6 **(a)** Photoelectric effect is the emission of electrons from a metal surface when it is